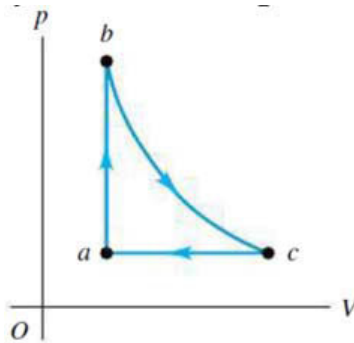


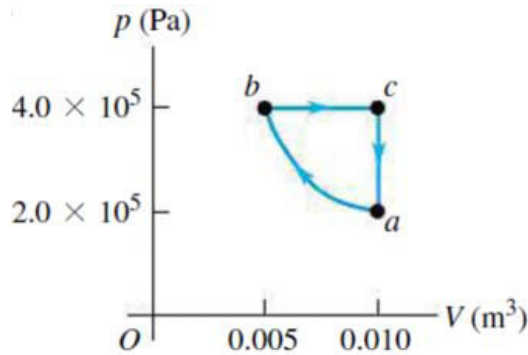
1. You are called upon to design a Carnot engine that has 2.00 mol of a monatomic ideal gas as its working substance and operates from a high temperature reservoir at  $500^{\circ}\text{C}$ . The engine is to lift a 15.0 kg weight 2.00 m per cycle, using 500 J of heat input. The gas in the engine chamber can have a minimum volume of 5.00 L during the cycle.
- What is the maximum pressure that the gas chamber will have to withstand?  $(2.5712 \times 10^6 \text{ N/m}^2)$
  - How much heat energy does this engine waste per cycle?  $(206 \text{ J})$
  - What is the thermal efficiency of the engine?  $(0.588)$
  - What must be the temperature of the cold reservoir?  $(318.54 \text{ K})$

2. A heat engine operates using the cycle shown in figure.

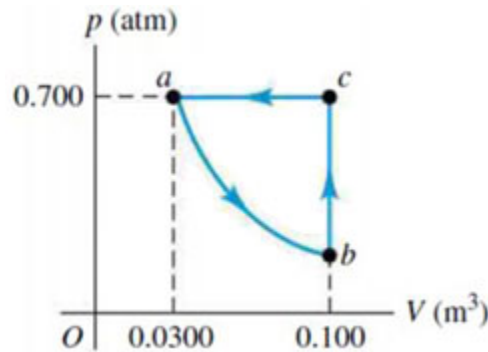


The working substance is 2.00 mol of helium gas, which reaches a maximum temperature of  $327^{\circ}\text{C}$ . Assume the helium can be treated as an ideal gas. Process  $bc$  is isothermal. The pressure in states  $a$  and  $c$  is  $1.00 \times 10^5 \text{ Pa}$ , and the pressure in state  $b$  is  $3.00 \times 10^5 \text{ Pa}$ .

- How much heat enters the gas and how much leaves the gas each cycle?  $(20944.72 \text{ J}, 16633.2 \text{ J})$
  - How much work does the engine do each cycle, and what is its efficiency?  $(0.2058)$
  - Compare this engine's efficiency with the maximum possible efficiency attainable with the hot and cold reservoirs used in this cycle.  $(0.3087)$
3. You build a heat engine that takes 1.20 mol of an ideal diatomic gas through the cycle shown in figure.



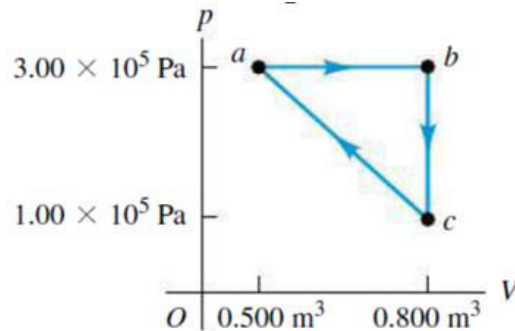
- (i) Show that process  $ab$  is an isothermal compression.
  - (ii) During which process(es) of the cycle is heat absorbed by the gas? During which process(es) is heat rejected? How do you know?
  - (iii) Calculate the temperatures at points  $a$ ,  $b$ , and  $c$ . (200.47 K, 200.47 K, 400.93 K)
  - (iv) Calculate the net heat exchanged with the surroundings and net work done by the engine in one cycle. (613.71 J, 613.71 J)
  - (v) Calculate the thermal efficiency of the engine. (0.08767)
  - (vi) Compare this engine's efficiency with the maximum possible efficiency attainable with the hot and cold reservoirs used in this cycle. (0.17534)
4. Prove that the reverse Carnot engine delivers the best theoretical performance for refrigeration. (Hint: you can prove it by contradiction.)
5. The  $pV$ -diagram in the figure shows the cycle for a refrigerator operating on 0.850 mol of  $H_2$ . Assume that the gas can be treated as ideal. Process  $ab$  is isothermal.



- (i) Find the coefficient of performance of this refrigerator. (6.23)
  - (ii) Compare this refrigerator's coefficient of performance with the maximum possible coefficient of performance attainable with the cold and hot reservoirs used in this cycle.
6. You make tea with 0.250 kg of  $88.0^\circ\text{C}$  water and let it cool to room temperature ( $20.0^\circ\text{C}$ ). The specific heat of tea water is  $4190 \text{ J/kg}\cdot^\circ\text{C}$ .
- (i) Calculate the entropy change of the water while it cools. ( $-218.52 \text{ JK}^{-1}$ )
  - (ii) The cooling process is essentially isothermal for the air in your kitchen. Calculate the change in entropy of the air while the tea cools, assuming that all of the heat lost by the water goes into the air. ( $242.98 \text{ JK}^{-1}$ )

(iii) What is the total entropy change of the system of tea and air?  $(24.36 \text{ JK}^{-1})$

7. A monatomic ideal gas is taken around the cycle in the direction shown in figure. The path for process  $c \rightarrow a$  is a straight line in the  $pV$ -diagram.



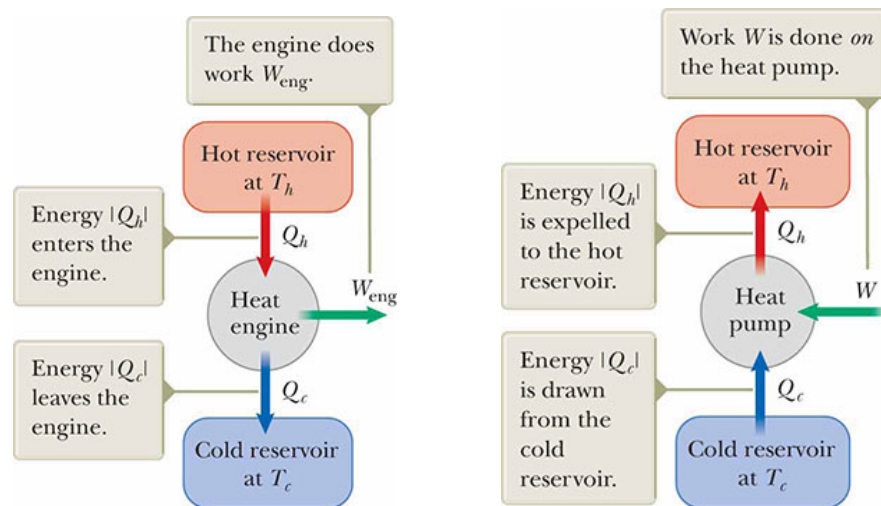
- (i) Calculate  $Q$ ,  $W$ , and  $\Delta U$  for each process  $a \rightarrow b$ ,  $b \rightarrow c$ , and  $c \rightarrow a$ .  
 $(2.25 \times 10^5 \text{ J}, 9 \times 10^4 \text{ J}, 1.35 \times 10^5 \text{ J}), (-2.4 \times 10^5 \text{ J}, 0, -2.4 \times 10^5 \text{ J}), (0.45 \times 10^5 \text{ J}, -0.6 \times 10^5 \text{ J}, 1.05 \times 10^5 \text{ J})$ .
  - (ii) What are  $Q$ ,  $W$ , and  $\Delta U$  for one complete cycle?  $(3 \times 10^4 \text{ J}, 3 \times 10^4 \text{ J}, 0)$
  - (iii) What is the efficiency of the cycle?  $\left(\frac{1}{9}\right)$
  - (iv) Compare this engine's efficiency with the maximum possible efficiency attainable with the hot and cold reservoirs used in this cycle.  $\left(\frac{1}{6}\right)$
8. During each cycle, a refrigerator ejects 625 kJ of energy to a high-temperature reservoir, and takes in 550 kJ of energy from a low-temperature reservoir.
- (i) Determine the work done on the refrigerant in each cycle.  $(75 \text{ kJ})$
  - (ii) Determine the coefficient of performance of the refrigerator.  $(7.333)$
9. During a cold day you have a clever idea to convert your air-conditioner at the window of your bedroom as a heating assembly by re-arranging the air-conditional 180° so that your bedroom will be warm and comfortable.



The original arrangement of the air-conditioner in your bedroom

The coefficient of performance (COP) for this heating assembly used to warm up a room during cold winter is modeled by 2 heat reservoirs  $Q_{\text{Hot}}$  and  $Q_{\text{Cold}}$ , and the work  $W$  done by the compressor in the heating assembly, and is defined as  $\text{COP}_{\text{heating assembly}} = \frac{Q_{\text{Hot}}}{W}$ . By conservation of energy,  $Q_{\text{Hot}} = Q_{\text{Cold}} + W$ . If  $\text{COP}_{\text{heating assembly}} = 4.2$ , and the compressor requires a power of 1.75 kW to operate,

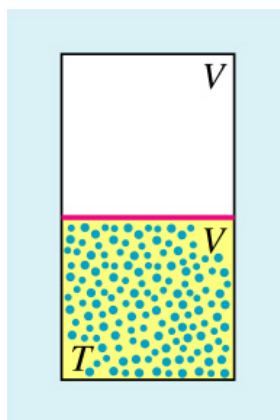
- (i) how much energy does this reversed air-conditioner add to your room in one hour?  
( $2.646 \times 10^7 \text{ J}$ )
  - (ii) if the reversed air-conditioner is reversed again so that it works as the original air conditioner in the hot summer, what would be its coefficient of performance? (3.2)
10. If a 35%-efficient Carnot heat engine (left figure) is run in reverse so as to form a refrigerator (right figure), what would be this refrigerator's coefficient of performance in terms of  $T_h$  and  $T_c$  ?



11. A heat engine is being designed to have a Carnot efficiency of 65.0% when operating between two heat reservoirs.
  - (i) If the temperature of the cold reservoir is  $20.0^\circ\text{C}$ , what must be the temperature of the hot reservoir? ( $564.42^\circ\text{C}$ )
  - (ii) Can the actual efficiency of the engine be equal to 65%? Explain your answer.
12. A sample of an ideal gas expands isothermally, doubling in volume.
  - (i) Write down the work done on the gas in this expansion.
  - (ii) Because the internal energy of an ideal gas depends solely on its temperature, the change in internal energy is zero during the isothermal expansion. It follows from the first law of thermodynamic that the energy input to the gas by heat during the expansion is equal to the energy output by work. Does this process have 100% efficiency in converting energy input by heat into work output?
  - (iii) Does the conversion in (ii) violate the second law of thermodynamics? Explain.

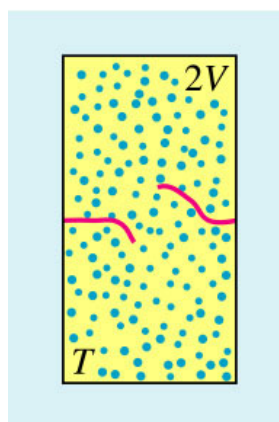
13. A sample consisting of  $n$  moles of an ideal gas undergoes a reversible isobaric expansion from volume  $V_i$  to  $3V_i$ . What is the change in entropy of the gas?

14. A thermally insulated box is divided by a partition into two compartments, each having volume  $V$ . Initially, one compartment contains  $n$  moles of an ideal gas at temperature  $T$ , and the other compartment is evacuated as shown in figure.



We then break the partition, and the gas expands to fill both compartments as shown in figure.

What is the entropy change in this free-expansion process? ( $nR \ln 2$ )



- End -

## Formula List

### Logarithm:

$$\log a + \log b = \log(a \times b)$$

$$\log a - \log b = \log\left(\frac{a}{b}\right)$$

$$(\log x) > 0, \text{ if } x > 1.$$

$$(\log 1) = 0.$$

$$(\log x) < 0, \text{ if } 0 < x < 1.$$

$$(\log x) \text{ is undefined if } x < 0.$$

### Trigonometry:

$$\sin = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sin}{\cos}$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\cos 2\theta = 1 - 2 \sin^2 \theta$$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

### Differentiation:

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(e^{f(x)}) = e^{f(x)} \times \frac{d}{dx}(f(x))$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(f(x) \times g(x)) = f(x) \times \left(\frac{d}{dx}g(x)\right) + \left(\frac{d}{dx}f(x)\right) \times g(x)$$

### Integration:

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + c$$

### Vector:

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}$$

$$\vec{A} \bullet \vec{B} = |\vec{A}| \times |\vec{B}| \times \cos \theta$$

### Linear kinematic equations:

$$v_f = v_i + at$$

$$\Delta x = v_i t + \frac{1}{2} at^2$$

$$\Delta x = \frac{(v_i + v_f)}{2} \times t$$

$$v_f^2 = v_i^2 + 2a\Delta x$$

### Simple harmonic motion:

$$x(t) = r_0 \cos(\theta_0 + \omega_0 t)$$

$$v_x(t) = -r_0 \omega_0 \sin(\theta_0 + \omega_0 t)$$

$$a_x(t) = -r_0 \omega_0^2 \cos(\theta_0 + \omega_0 t) \\ = -\omega_0^2 x(t)$$

$$\omega^2 = \frac{k}{m}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$v_{\max} = \sqrt{\frac{k}{m}} A$$

$$a_{\max} = \frac{k}{m} A$$

### Retarding force:

$$R = -bv$$

**Projectile motion:**

$$y = -\frac{g}{2v_{0x}^2}(x - x_0)^2 + \frac{v_{0y}}{v_{0x}}(x - x_0) + y_0, \text{ or}$$

$$y = x \tan \theta - \frac{gx^2}{2v_0^2}(1 + \tan^2 \theta)$$

**Uniform circular motion:**

$$\theta = \frac{\text{arc length}}{\text{radius}} = \frac{s}{r}$$

$$\omega = \frac{d\theta}{dt} = \frac{2\pi}{T}$$

$$v = r\omega$$

$$a_c = r\omega^2 = \frac{v^2}{r}$$

**Non-uniform circular motion:**

$$a_t = \frac{dv}{dt}$$

$$a = \sqrt{a_c^2 + a_t^2}$$

**Relative velocity:**

$$\vec{V}_{P/A} = \vec{V}_{P/B} + \vec{V}_{B/A}$$

**Newton's second law:**

$$\sum \vec{F} = m\vec{a}$$

**Newton's third law:**

$$\vec{F}_{12} = -\vec{F}_{21}, \text{ and the two forces} \\ \text{act on the opposite body.}$$

**Frictional force:**

$$f_s \leq \mu_s n, \text{ and impending motion} \\ \text{occurs when } f_s = \mu_s n.$$

$$f_k = \mu_k n$$

**Translational kinetic energy:**

$$KE_{\text{translational}} = \frac{1}{2}mv^2$$

**Gravitational potential energy:**

$$\Delta PE = mg(y_2 - y_1) \\ = mgh$$

**Spring**

$$F_{\text{spring}} = -kx$$

$$\text{Elastic } PE_{\text{spring}} = \frac{1}{2}kx^2$$

**Work-energy theorem:**

$$W = \Delta KE = KE_f - KE_i$$

**Work done (constant force):**

$$W = \vec{F} \bullet \Delta \vec{r} = F \Delta r \cos \theta$$

**Work done (varying force):**

$$W = \int_{x_i}^{x_f} \vec{F} \bullet d\vec{r}$$

**Power:**

$$P = \frac{dW}{dt} = F \times v$$

**Linear momentum:**

(also valid for y and z)

$$p_x = mv_x$$

$$F_x = \frac{dp_x}{dt}$$

$$F_{\text{average, x}} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} F_x(t) dt$$

**Conservation of linear momentum:**

$$\vec{p}_{1,i} + \vec{p}_{2,i} = \vec{p}_{1,f} + \vec{p}_{2,f}$$

**Elastic collision:**

$$v_{A1} - v_{B1} = -(v_{A2} - v_{B2})$$

$$v_{A2} = \left( \frac{m_A - m_B}{m_A + m_B} \right) v_{A1} + \left( \frac{2m_B}{m_A + m_B} \right) v_{B1}$$

$$v_{B2} = \left( \frac{m_B - m_A}{m_A + m_B} \right) v_{B1} + \left( \frac{2m_A}{m_A + m_B} \right) v_{A1}$$

**Center of mass**

(also valid for y and z)

$$x_{\text{cm}} = \frac{1}{M} \sum m x, \text{ where } M = \sum m, \text{ or}$$

$$= \frac{1}{M} \int x \, dm$$

$$v_{\text{cm},x} = \frac{1}{M} \sum m v_x$$

$$a_{\text{cm},x} = \frac{1}{M} \sum m a_x$$

**Simple Harmonic Motion:**

$$a_x = -\omega^2 x$$

$$\omega^2 = \frac{k}{m}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$v_{\text{max}} = \sqrt{\frac{k}{m}} A$$

$$a_{\text{max}} = \frac{k}{m} A$$

**Retarding force:**

$$F = -b v$$

**Rotation**

$$\theta = \frac{s}{r}$$

$$\omega = \frac{d\theta}{dt}$$

$$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$

$$v_{\text{tangential}} = r\omega$$

$$a_{\text{centripetal}} = r\omega^2 = \frac{v^2}{r}$$

$$a_{\text{tangential}} = r\alpha$$

**Rotational kinematic equations:**

$$\omega_f = \omega_i + \alpha t$$

$$\Delta\theta = \omega_i t + \frac{1}{2} \alpha t^2$$

$$\Delta\theta = \frac{(\omega_i + \omega_f)}{2} \times t$$

$$\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$$

**Moment of inertia:**

$$I = \sum m r^2$$

$$I = \int r^2 \, dm$$

**Parallel axis theorem**

$$I = I_{\text{cm}} + M d^2$$

**Rotational kinetic energy:**

$$KE_{\text{rotational}} = \frac{1}{2} I \omega^2$$

**Moment of inertia:**

$$I = \sum m_i r_i^2$$

$$I = \int r^2 \, dm$$

**Moment of inertia of hoop or thin cylindrical shell:**

$$I_{\text{CM}} = MR^2$$

**Moment of inertia of hollow cylinder:**

$$I_{\text{CM}} = \frac{1}{2} M (R_1^2 + R_2^2)$$

**Moment of inertia of solid cylinder:**

$$I_{\text{CM}} = \frac{1}{2} MR^2$$

**Moment of inertia of rectangular plate:**

$$I_{\text{CM}} = \frac{1}{12} M (a^2 + b^2)$$



**Moment of inertia of thin rod about its CM:**

$$I_{CM} = \frac{1}{12} ML^2$$

**Moment of inertia of thin rod about its end:**

$$I = \frac{ML^2}{3}$$

**Moment of inertia of solid sphere:**

$$I_{CM} = \frac{2}{5} MR^2$$

**Moment of inertia of spherical shell:**

$$I_{CM} = \frac{2}{3} MR^2$$

**Torque and Moment of force:**

$$\tau = F \times r \times \sin \theta$$

$$\sum \tau = I\alpha$$

**Work due to rotation:**

$$\Delta W = \tau \times \Delta \theta$$

$$W = \int_{\theta_1}^{\theta_2} \tau d\theta$$

**Power due to rotation:**

$$P = \tau \times \omega$$

**Pure rolling:**

$$v_{cm} = r\omega$$

$$a_{cm} = r\alpha$$

$$KE_{\text{total}} = \frac{1}{2} Mv_{cm}^2 + \frac{1}{2} I_{cm} \omega^2$$

**Conservation of Angular momentum:**

$$\underbrace{\sum I\omega}_{\text{Before collision}} = \underbrace{\sum I\omega}_{\text{After collision}}$$

**Avogadro's number:**

$$N_A \approx 6.022 \times 10^{23} \text{ particles/mole}$$

**Ideal gas constant:**

$$R \approx 8.314 \text{ J mole}^{-1} \text{ K}^{-1}$$

**Boltzmann constant:**

$$k = \frac{R}{N_A} \approx 1.381 \times 10^{-23} \text{ J/K}$$

**Ideal gas equation:**

$$pV = nRT$$

**Speed of a molecule**

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

**Mole**

$$n = \frac{N}{N_A}$$

$$n = \frac{m}{M}$$

**Absolute temperature**

$$T = T_{\text{Celsius}} + 273.15$$

**Specific heat  $c$  and molar heat  $C$**

$$Q = Mc(T_2 - T_1)$$

$$= nC(T_2 - T_1)$$

**Latent heat  $L$**

$$Q = L \times M$$

**First law of thermodynamics**

$$\Delta U = Q - W$$

**Internal energy for  $N$  molecules**

$$\text{monatomic: } U = N \times \frac{3}{2} kT$$

$$\text{diatomic: } U = N \times \frac{5}{2} kT$$

**Relationship between  $C_p$  and  $C_v$  for an ideal Gas**

$$C_p - C_v = R$$

**Isochoric process**

$$\text{monatomic: } C_V = \frac{3}{2}R$$

$$\text{diatomic: } C_V = \frac{5}{2}R$$

$C_p$  is undefined for isochoric process.

**Isobaric process**

$$\text{monatomic: } C_p = \frac{5}{2}R$$

$$\text{diatomic: } C_p = \frac{7}{2}R$$

$C_V$  is undefined for isobaric process.

**Isothermal process**

$$W = nRT \ln \left( \frac{V_f}{V_i} \right)$$

**Adiabatic process**

$$\gamma = \frac{C_p}{C_V}$$

$$\text{monatomic: } \gamma = \frac{5}{3}$$

$$\text{diatomic: } \gamma = \frac{7}{5}$$

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$P_1^{1-\gamma} T_1^\gamma = P_2^{1-\gamma} T_2^\gamma$$

**Thermal efficiency of a heat engine**

$$e = 1 - \frac{|Q_C|}{|Q_H|}$$

**Coefficient of performance of a heat pump for cooling**

$$\text{COP} = \frac{|Q_C|}{|Q_H| - |Q_C|}$$

**Entropy**

$$\Delta S = \int_{\text{initial state}}^{\text{final state}} \frac{dQ}{T} \geq 0$$

**Coulomb's Law:**

$$F = \frac{kq_1 q_2}{r^2}$$

where

$$k = 8.9875 \times 10^9 \text{ Nm}^2/\text{C}^2$$

**Electric field:**

$$E = \frac{F}{q_0}$$

**Electric field for point charge:**

$$E = \frac{kq}{r^2}$$

**Electric field at a distance along central axis of charged ring:**

$$E = \frac{kQx}{(x^2 + a^2)^{3/2}}$$

**Electric potential:**

$$V = \frac{U}{q_0}$$

**Electric potential in uniform field:**

$$\Delta V = -Ed$$

**Electric potential for point charge:**

$$V = \frac{kq}{r}$$

**Electric potential at a distance along central axis of charged ring:**

$$V = \frac{kQ}{\sqrt{x^2 + a^2}}$$

**Electric potential energy (for multiple point charges):**

$$U = k \left( \frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right)$$

Electric current:

$$I = \frac{dQ}{dt}$$
$$I_{av} = nqv_d A$$

Ohm's Law:

$$I = \frac{\Delta V}{R}$$

Resistance of ohmic conductor:

$$R = \rho \frac{L}{A}$$

Resistivity:

$$\rho = \rho_0 [1 + \alpha(T - T_0)]$$

Electric power:

$$P = I\Delta V = I^2 R = \frac{\Delta V^2}{R}$$

Electric energy:

$$U = P\Delta t$$

Terminal voltage:

$$\Delta V = \mathcal{E} - Ir$$

Resistors in series:

$$R_{eq} = R_1 + R_2 + R_3 + \dots$$

Resistors in parallel:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

Kirchhoff's Junction Rule:

$$\sum I_{in} = \sum I_{out}$$

Kirchhoff's Loop Rule:

$$\sum \Delta V = 0$$

Magnetic force on a moving charge:

$$F_B = qvB \sin \theta$$

Magnetic force on a current carrying conductor:

$$F = IlB \sin \theta$$

Radius of moving charged particle in uniform magnetic field:

$$r = \frac{mv}{qB}$$

Angular speed:

$$\omega = \frac{qB}{m}$$

Period:

$$T = \frac{2\pi m}{qB}$$

Velocity selector:

$$v = \frac{E}{B}$$

Magnetic field due to long straight conductor:

$$B = \frac{\mu_0 i}{2\pi a} \text{ where}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$$

Magnetic field at center of circular wire loop:

$$B = \frac{\mu_0 I}{2R}$$

Magnetic field at a distance along central axis of current-carrying ring:

$$B = \frac{\mu_0 IR^2}{2(x^2 + R^2)^{3/2}}$$

Magnetic force between parallel conductors:

$$F = \frac{\mu_0 I_1 I_2 l}{2\pi a}$$

Magnetic field inside toroid:

$$B = \frac{\mu_0 NI}{2\pi r}$$

Magnetic field inside solenoid:

$$B = \mu_0 nI$$

**Magnetic flux:**

$$\Phi_B = BA \cos \theta$$

**Faraday's Law:**

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}$$

**Motional emf:**

$$\mathcal{E} = \frac{Blv}{R}$$

**Power input:**

$$P = F_{\text{app}} v = I l B v$$